

A Comparative Study of Machine Learning Algorithms for Customer Attrition Prediction in Banking Sector

Deepak Yadav

School of Computing Information

National College of Ireland

Dublin, Ireland

x21219991@student.ncirl.ie

Abstract—The objective of this study was to evaluate the performance of five different classifiers in predicting customer churn in the banking sector, using three datasets with the target feature “Churn”. The classifiers evaluated were Random Forest, Logistic Regression, Decision Tree, KNN, and ADABOOST. Performance metrics such as precision, recall, F1 score, ROC curve, and AUC score were used to compare the classifiers. The results showed that the Random Forest classifier outperformed the other classifiers in all three datasets, with an average AUC score of 0.86. Additionally, the study proposed a conceptual framework for churn management and highlighted the potential benefits of using innovative analytical approaches to improve customer retention. The findings of this study can be useful for banks seeking to enhance their churn management practices and improve customer satisfaction.

Index Terms—Customer engagement, banking services, data mining, Predictive modeling, Customer attrition

I. INTRODUCTION

According to McKinsey’s Global Banking Annual Review 2022, the number of consumers with banks and financial institutions has grown during the last decade and a half [1]. Banks offer services through a variety of channels, including ATMs, debit cards, credit cards, internet banking, and so on [2]. The massive growth in customer numbers has made banks more sensitive to the quality of service they provide [3]. This occurrence sparked fierce competition among numerous banks. In result of that I can see boost in bank services [4]. Customers changing from one bank to another has become a typical occurrence [5] [6]. It is due to factors such as new mobile app feature, customer-friendly bank employees, less interest rates, geographic closeness, the variety of services supplied, and so on [7]. Consequently, this switching has negative consequences for the bank, such as decreased profits and other operational problems. Furthermore, due to the rapid development of the digital age and cloud computing, the volume of data available has grown exponentially. The availability of information affects business models of organizations, particularly those of financial institutions. In order to mitigate customer churn, it is essential to identify and predict which clients are more likely to terminate their relationship with the company, then marketing efforts shall be conducting

for retaining them. As a result, a framework approach to identifying and modeling to guarantee client lifetime value is required, as is a deeper understanding of churn management.

The goal of this research is to investigate the application of machine learning algorithms in managing bank customer churn. Several machine learning algorithms shall be applied for comparative analysis and decision-making using the output model. The results of this research will have significant implications for the financial sector’s ability to retain its customers.

Built on a thorough investigation of the literature and business practices that have been reviewed with association to exit analysis and to create a conceptual framework for exit management. The research’s precise goals are as follows:

- To present a machine learning method for churn prediction innovation.
- How the employment of innovative analytical approaches enhances the decision-making process for customer churn management ?
- To compare five machine learning classifiers on three different datasets for churn prediction.
- Which Machine Learning techniques are conducive for retention customer’s campaign.

The results of this study will be beneficial for developing better churn management practices in the banking sector.

II. LITERATURE REVIEW

Churn prediction has attracted considerable interest in the past ten years, especially in the following industries: Electric Commercial [8], Banking [7], Credit Card [9], Digital Game [10], Insurance [11], Telecommunication [12], Health care [13], Social media [14], Energy [15]. This discussion has shown that a variety of enterprises must address the issue of client churn. Table I gives a quick summary of earlier studies in the banking sector.

Due to their ability to combine strong predictive accuracy with good interpretation, decision trees (DT) and logistic regression (LR) are two of the most widely used techniques for predicting a churn likelihood in customer churn prediction [9], [11]. While both approaches are beneficial and have their advantages, they also have drawbacks. DT is particularly good

TABLE I
RECENT STUDIES ON CUSTOMER ATTRITION

Title	Techniques	Evaluation	Result
A Survey on Customer Churn Prediction using ML Techniques (2016) [16]	Artificial neural networks (ANN), Decision trees, and Logistic regression	Accuracy, precision, recall, and F1-score, as well as confusion matrix and ROC curve analysis	A literature survey
Churn prediction using comprehensible support vector machine: An analytical CRM application (2014) [17]	SVM, NBTree, Over-sampling, Under-sampling	T-test, sensitivity, specificity and Accuracy	SVM + NBTree got the best sensitivity of 91.85%
Machine Learning Based Customer Churn Prediction In Banking (2020) [18]	SVM, Minimum Redundancy Maximum Relevance (mRMR), Decision Tree, Random Forest	Accuracy, AUC	Random Forest with 92.95% accuracy.
Churn Prediction in Banking using ML with ANN (2022) [7]	SVM, Decision tree, Random Forest, ANN	Accuracy, Precision, Recall	Random Forest with accuracy of 86%
Dynamic churn prediction framework with more effective use of rare event data: The case of private banking (2014) [19]	SPTD and MPTD	Classification accuracy and AUC	MPTD with 75% AUC
Client Churn Prediction of Banking and fund industry utilizing Machine Learning Techniques (2019) [20]	Decision Tree and Neural Networks	Accuracy, Recall, Precision and F1-Score	Neural Networks achieved 93% of Recall and 85% of Accuracy.
Credit card churn forecasting by logistic regression and decision tree (2011) [9]	Logistic Regression and Decision Tree	Type I, II errors, Misclassification cost	Logistic regression with lower misclassification
Prediction of Customer Attrition of Commercial Banks based on SVM Model (2014) [21]	Support Vector Machine (SVM)	F-measure, sensitivity, ROC curve	Logis SVM with 99.09%
Customer Churn Analysis and Prediction in Banking Industry using Machine Learning (2020) [22]	Logistic regression (LR), Decision tree (DT), K-nearest neighbor (KNN), Random forest (RF)	AUC and Accuracy	Random forest classifier with 85.22% accuracy
Modeling churn using customer lifetime value (2009) [23]	Logistic regression, Neural network, Decision tree, AdaCost, Cost-sensitive tree	TPR and AUC	AdaCost with 96.6% TPR and 91.94% of AUC
Customer Churn Analysis and Prediction Using Data Mining Models in Banking Industry (2019) [24]	Decision Tree, Neural Network, SVM, Naive Bayes and Logistics Regression	Accuracy, AUC, Precision and Recall	SVM with 92.65% and 75% of AUC.
A novel hybrid undersampling method for mining unbalanced datasets in banking and insurance (2015) [11]	Decision Tree (DT), Support Vector Machine (SVM), Logistic Regression (LR), Probabilistic Neural Network (PNN), Group Method of Data Handling (GMDH), Multi-Layer Perceptron (MLP)	Sensitivity, Specificity, Accuracy and AUC	Decision Tree produced the best sensitivity of 91.2%

at handling interactions between variables, but it struggles with handling linear relationships. Contrarily, LR does not discover and account for inter-variable interaction effects; instead, it handles linear interactions between factors quite effectively. Additionally, Random Forest performed quite well in numerous studies on churn in banking [18] and [22], it even superior to a novel ANN in [7]. In studies of [23] [17] and [24] Adaboost and SVM also worked well, then these two techniques also serve as wise options for churn research. Also based on literature review, AUC and accuracy metrics are common measures in the assessment of forecasting models as well as churn prediction.

III. DATA AND METHODOLOGY

A. Customer Churn

A churn client in terms of financial services is someone who stops transacting with a bank and closes all of their banking accounts. Client turnover can negatively impact a company's

operations and cause money and earnings to lose value. Therefore, one essential component of customer-centricity retention, which aims to reduce churn, is finding and identifying clients who show a strong tendency to leave the company or predict client churn.

B. Dataset and Business Understanding

Three datasets used in this study are collected from Kaggle which is a Data Science and Machine Learning platform under a subsidiary of Google LLC.

All chosen three datasets have the target feature ("Churn", "Exited" and "Attrition_Flag"), which indicates whether the customer has left, indicated by (Yes/1/Attrited Customer) or (No/0/Existing Customer) respectively.

XYZ Bank Customer Churn Dataset: The "XYZ Bank Customer Churn" dataset is a collection of 10,000 observations containing 14 customer-related features and can be used to analyze customer behavior and predict churn. It was obtained from Kaggle [25] and includes Geography, Age, Gender,

TABLE II
DATASET OVERVIEW

Description	ABC Bank	BNM Bank	XYZ Bank
Number of observations	28,382	10,127	10,000
Number of features	21	13	14
Number of classes	2	2	2
Number of churn	5,260	1,627	2,037
Number of non-churn	23,122	8,500	7,963
Percentage of churn	18.5%	16.1%	20.37%
Percentage of non-churn	81.5%	83.9%	79.63%

financial information such as credit score and account balance, as well as customer behavior attributes such as associated No. of products and whether they have a credit card. Exited column displays if a customer is no longer with the bank or not. This dataset has been previously utilized in churn analysis research within the banking industry [24] [22].

ABC Bank Customer Churn Dataset: The dataset contains customer data from a bank, including demographic information, account details, and transaction history. It includes 20 columns such as age, gender, current balance, average monthly balance, and churn. There are no missing values in the dataset. The dataset consists of 28382 rows, and the data ranges from January 2004 to September 2016 [26]. The dataset appeared in the researchers of [22] [27] for customer attrition risk prediction in retail banking.

BNM Bank Customer Churn Dataset: The BNM Bank Customer Churn is a dataset available on Kaggle [28], containing information about customers and their banking activities. The dataset comprises 10,000 records and 21 columns. It includes customer ID, gender, age, income, education level, credit card utilization, and churn status. The dataset has been derived to include columns like average utilization and balance to help analyze the behavior of churned customers. The primary objective of the dataset is to help the bank understand factors contributing to customer churn and improve its customer retention efforts. The dataset can be utilized for building predictive models to identify at-risk customers and performing exploratory data analysis to identify patterns and trends in customer behavior. It is also one of the research subjects in churn prediction [22].

C. Techniques

Numerous Data Mining techniques have been employed in academic research to forecast customer churn in the banking industry from 2009 to 2022. These techniques include Decision Trees, Logistic Regression, KNN, SVM, Random Forest as displayed below:

- **Random Forest:** Classification tree collection is used in Random Forest. Each tree is constructed with the class variable on its leaf nodes and dependent variables on the intermediate nodes. When a new data point is inputted into the model, each tree in the forest provides its own classification, and the final classification is determined by the category with the highest number of votes from all the trees in the forest [29].

- **Logistic Regression (LR):** A machine learning approach utilized to estimate the likelihood of an occurrence based on independent variables. It calculates the log function to generate numerical output ranging from 0 to 1. It is commonly employed across different domains to tackle binary classification problems [30].
- **Decision Tree** is used for classification and regression tasks [31]. The algorithm recursively divides the data based on the feature values until it reaches a stopping point. Decision trees are easy to interpret and can handle missing values. However, they may also experience overfitting issues [32].
- **K-nearest Neighbor (KNN)** is a technique used for regression and classification tasks. It predicts the outcome of a new data point by examining the k-nearest neighbors' values or classes in the training data. This approach is uncomplicated to implement and effective for dealing with both categorical and numerical data. Nevertheless, it can be computationally expensive and dependent on the selection of k and distance metric [32].
- **AdaBoost** [33] is an ensemble learning technique that combines multiple weak classifiers to form a strong classifier. In the case of classification problems, a weak classifier is one that performs only slightly better than random guessing, such as a decision tree with limited depth or a simple linear classifier. AdaBoost works by iteratively training these weak classifiers on different subsets of the training data, assigning weights to each data point based on how well the previous classifiers performed on that point, and then combining the classifiers using a weighted majority vote. The idea behind AdaBoost is that by reweighting the data and focusing on the misclassified examples, subsequent classifiers will be better able to correct the errors of previous classifiers. This process continues until a predefined number of classifiers has been trained or until the desired level of accuracy has been reached. AdaBoost can be used with many different base classifiers, including decision trees, neural networks, and support vector machines. In the paper mentioned earlier, AdaBoost is used in conjunction with Random Forest Classifier, Decision Trees Classifier, and Extra Tree Classifier to predict customer churn in a telecommunications dataset. The combination of these classifiers through AdaBoost resulted in improved prediction accuracy compared to using any single classifier alone. [33].

D. Performance measures

For binary classification problem of predicting customer churn, the confusion matrix is a helpful tool to evaluate performance, providing fmy results: TP, TN, FP, and FN [34]. Furthermore, Precision, Recall and Specificity, F1 score or F-measure are commonly derived from the confusion matrix for performance assessment. Comparing models visually can be achieved through the use of AUC-ROC [35] [36].

An explanation of the technical formulations and meanings of the indicators is based on the confusion matrix in Figure 1

		Predicted Class		
		Positive	Negative	
Actual Class	Positive	True Positive (TP)	False Negative (FN) Type II Error	Sensitivity $\frac{TP}{(TP + FN)}$
	Negative	False Positive (FP) Type I Error	True Negative (TN)	Specificity $\frac{TN}{(TN + FP)}$
		Precision $\frac{TP}{(TP + FP)}$	Negative Predictive Value $\frac{TN}{(TN + FN)}$	Accuracy $\frac{TP + TN}{(TP + TN + FP + FN)}$

Fig. 1. Confusion matrix with advanced classification metrics

Accuracy is defined as the ratio of the number of correct predictions to the total number of predictions made by the model [37].

Precision is defined as the ratio of the number of true positives to the total number of positive predictions made by the model [38].

Recall/Sensitivity is defined as the ratio of the number of true positives to the total number of actual positive instances in the dataset [38].

F1 score is the harmonic mean of precision and recall. It provides a single score that balances both precision and recall [38].

ROC Curve A plot that shows the performance of a binary classification model at different classification thresholds [39].

AUC Score The area under the ROC curve. It provides a single score that summarizes the performance of a binary classification model at all classification thresholds [39].

This study also uses the stratified K-fold cross-validation method due to the significant data imbalance [40].

E. Machine Learning pipeline

The CRISP DM model process is used in this project [41], which is widely recognized as the standard and neutral process model for executing data mining initiatives across industries.

As shown in the Fig 2, the CRISP-DM framework is built on a six-phase process that includes Business understanding, Data understanding, Data preparation, Modeling, Evaluation, and Deployment.

IV. EXPERIMENTS

A. Business and Data Understanding

Based on the analysis presented in Figures 3, 4, and 5, it can be observed that the BNM Bank dataset has a lower proportion of churned customers (16.1%) as compared to the proportion of existing customers (83.9%). On the other hand, the ABC

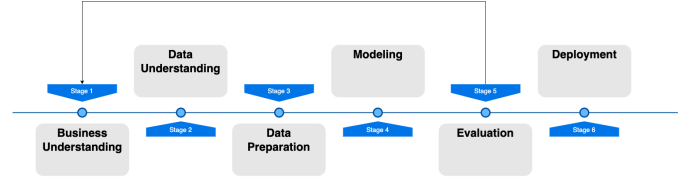


Fig. 2. Standard supervised machine learning pipeline

Bank dataset has a higher percentage of churning customers (81.5%) compared to non-churning customers (18.5%). Similarly, in the XYZ Bank dataset, the percentage of churning customers is 20.37%, while non-churning customers account for 79.63% of the dataset.

These ratios provide an initial understanding of the frequency at which churned and non-churned customers appear in each dataset. It is worth noting that the BNM Bank dataset stands out as having a significantly lower percentage of churned customers compared to the other two datasets.

The insights gained from this initial analysis is served as a basis for further exploration of the datasets. These findings inform the development of predictive models that can be used to identify potential churners and develop targeted retention strategies. In conclusion, these preliminary results provide a valuable foundation for analysis and modeling efforts aimed at reducing churn rates in the banking industry.

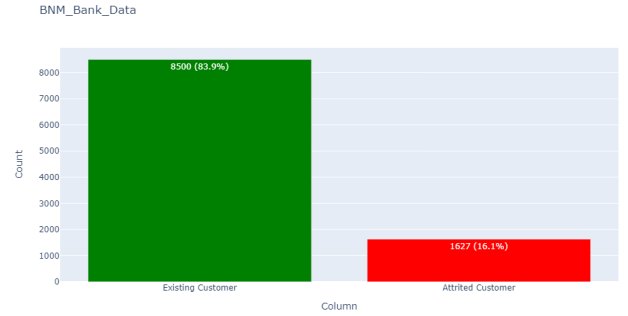


Fig. 3. Target variable percentage of the BNM Bank data

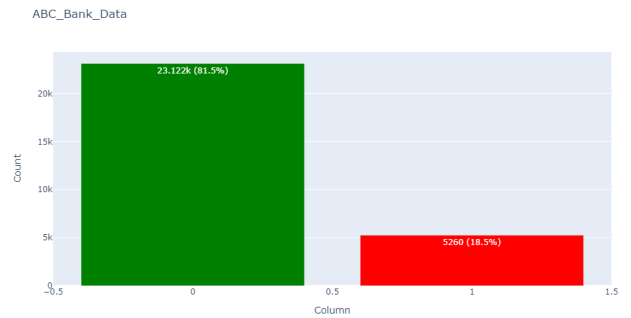


Fig. 4. Target variable percentage of the ABC Bank data

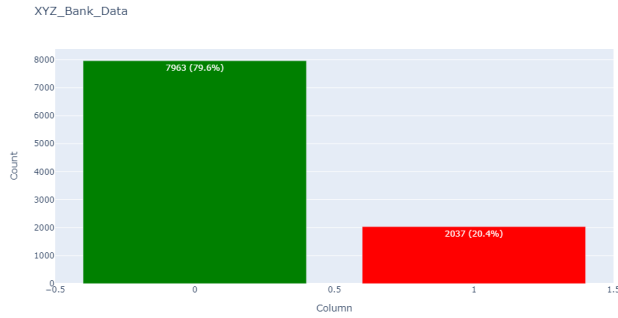


Fig. 5. Target variable percentage of the XYZ Bank data

B. Data Pre-processing

1) **Handling Missing Data:** Before I start with any data analysis, it is important to check if my dataset contains any missing or NA values. If there are any, I need to handle them appropriately to avoid any biases in my analysis.

In my study, I found missing data in some of the columns of the datasets:

- **BNM Bank:**

The columns *Education_Level* and *Marital_Status* contain 1519 and 749 missing rows respectively.

- **ABC Bank:**

The columns *gender*, *dependents*, *occupation*, and *city* contain 525, 2463, 80, and 803 missing rows respectively.

- **XYZ Bank:**

No missing or NA values were found in the dataset.

To handle missing data, I can either remove the rows with missing values or impute them with some other values. The choice of method depends on the amount and nature of missing data and the analysis to be performed. It is important to handle missing data appropriately to avoid any biases or inaccuracies in the analysis.

2) **Data Cleaning:** Data cleaning is an important step in any machine learning project. It involves identifying and correcting inaccurate, incomplete, or irrelevant data. In my study, I have identified some columns in my datasets that are not needed for further model making. I have also addressed missing or NaN values in the datasets. Here is a breakdown of what I did for each dataset:

- **BNM Bank:** I identified that the columns *Education_Level* and *Marital_Status* have 1519 and 749 rows of missing data, respectively. I replaced the **nan** values with **NaN** and used a categorical imputer to fill in the missing values in these columns. I also replaced any *abc* values in the *Income_Category* column with **NaN** and used the categorical imputer to fill in the missing values. I converted all categorical columns to strings and dropped the *CLIENTNUM* column, as it was not needed for further model making.

- **ABC Bank:** I dropped the *customer_id* and *last_transaction* columns, as they were not needed for further model making.

- **XYZ Bank:** I dropped the *RowNumber*, *CustomerId*, and *Surname* columns, as they were not needed for further model making.

Overall, data cleaning helps to ensure that my datasets are ready for analysis and modeling, and that I can obtain accurate and meaningful insights from my data.

C. Data Transformation

In this section, I shall be discussing the data transformation process for three different banks, namely BNM Bank, ABC Bank, and XYZ Bank.

- **BNM Bank:**

I first selected only the numerical columns using the `select_dtypes()` function and appropriate data types. Next, I calculated the skew and kurtosis for each column using the `agg()` and `transpose()` functions. If I found that any columns had a skew greater than zero, I transformed the data using appropriate techniques such as log transformation. Additionally, I removed any outliers from the data using appropriate techniques such as filtering by a certain threshold value. Finally, I replaced values in the *Attrition_Flag* column using the `replace()` function and converted the *Gender* column values to 0 or 1 using the `replace()` function.

- **ABC Bank:**

The first step was to replace missing values. I used the `SimpleImputer()` function to replace any missing numerical values with the median value, and missing categorical values with the most frequent value. Next, I changed the data types of certain columns such as *gender*, *dependents*, *occupation*, and *city* using the `astype()` function. Finally, I replaced the *gender* column values with 0 or 1 using the `replace()` function.

- **XYZ Bank:**

I simply replace the “Gender” column values with 0 or 1 using the `replace()` function.

D. Correlation matrix

A correlation matrix is a useful tool for analyzing the relationship between multiple variables in a dataset. It displays the pairwise correlations between all the variables in a matrix format. The matrix can range from a 2x2 for two variables to an nxn matrix for n number of variables. Correlation coefficients range from -1 to 1, with values closer to -1 or 1 indicating a strong correlation, while values closer to 0 indicate a weak correlation. A correlation coefficient of 0 indicates no correlation [42].

A correlation matrix can be used to identify which variables are strongly correlated, which can help in feature selection and reduce the dimensionality of the dataset. It can also help in identifying potential multicollinearity issues in regression analysis. In addition, it can be used to generate insights and hypotheses for further analysis [43].

- **BNM Bank:**

I first created a heatmap of the correlation between the numerical features using the Seaborn

library. I selected the following numerical columns to include in the correlation matrix: *Customer_Age*, *Dependent_count*, *Months_on_book*, *Total_Relationship_Count*, *Months_Inactive_12_mon*, *Contacts_Count_12_mon*, *Credit_Limit*, *Total_Revolving_Bal*, *Avg_Open_To_Buy*, *Total_Amt_Chng_Q4_Q1*, *Total_Trans_Amt*, *Total_Trans_Ct*, *Total_Ct_Chng_Q4_Q1*, and *Avg_Utilization_Ratio*.

The resulting plot showed in figure 6 that several pairs of columns had a correlation above 70%, including *Customer_Age* and *Months_on_book*, *Avg_Open_To_Buy* and *totalCredit_Limit*, *Avg_Utilization_Ratio* and *Total_Revolving_Bal*, and *Total_Trans_Amt* and *Total_Trans_Ct*. I removed these columns using the `drop()` function. The resulting data frame had fewer columns and was better suited for further analysis.

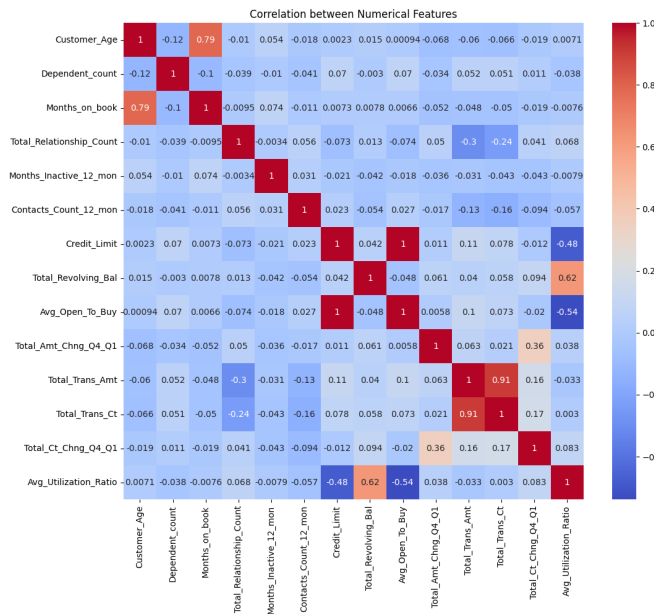


Fig. 6. Heatmap of the correlation between numerical features for BNM Bank

• ABC Bank:

I created a heatmap of the correlation between numerical features. The selected columns were: *current_balance*, *previous_month_end_balance*, *average_monthly_balance_prevQ*, *average_monthly_balance_prevQ2*, *previous_month_debit*, *previous_month_balance*, and *branch_code*. The resulting plot showed in figure 7 that several pairs of columns had a correlation above 70%. I removed these columns using the `drop()` function. The resulting data frame had fewer columns and was better suited for further analysis.

• XYZ Bank:

I created a heatmap of the correlation between numerical features. The selected columns were: *CreditScore*, *Gender*, *Age*, *Tenure*, *Balance*, *NumOfProducts*, *HasCrCard*, *IsActiveMember*, *EstimatedSalary*, and *Exited*. The

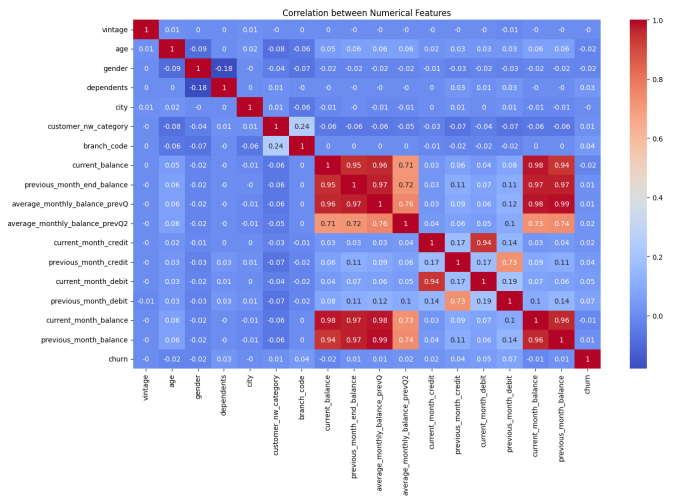


Fig. 7. Heatmap of the correlation between numerical features for ABC Bank

resulting plot showed Figure 8 that no pairs of columns had a correlation.

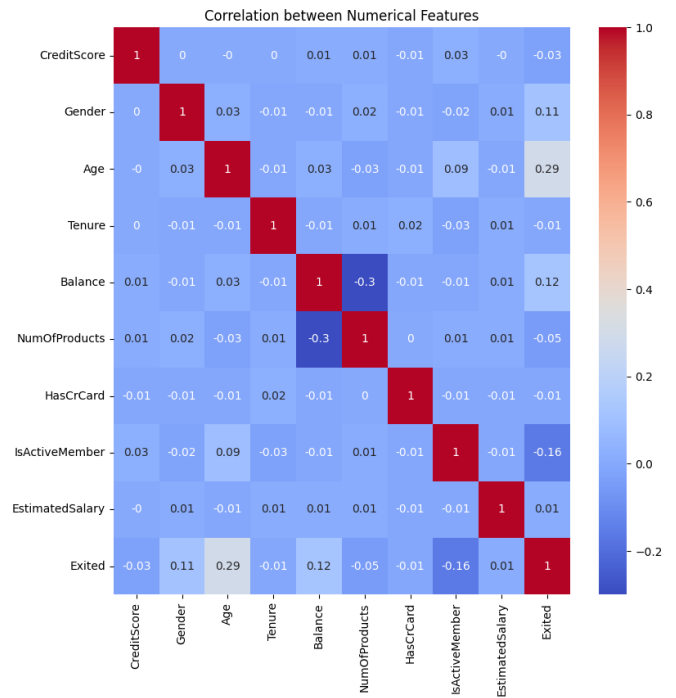


Fig. 8. Heatmap of the correlation between numerical features for XYZ Bank

E. Data Visualization

• BNM Bank:

For BNM Bank, I created the following data visualizations:

Customer age distribution by attrition flag: Stacked histogram of the customer age distribution based on their attrition flag figure 9 shows the distribution of customer ages, grouped by whether they have churned or not. This

helps us understand the age distribution of customers who are leaving the bank compared to those who are staying. We can see more number of churned customer age between 35 and 55.

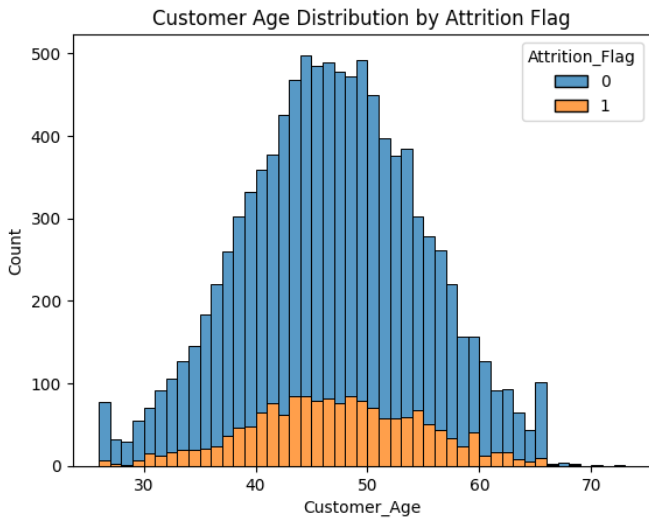


Fig. 9. Customer Age Distribution by Attrition Flag

Customer gender distribution by attrition flag: This chart in figure 10 shows the number of male (0) and female (1) customers who have churned or not. I observed that 17.4% Female customers have churned where only 14.6% male customers are churned.

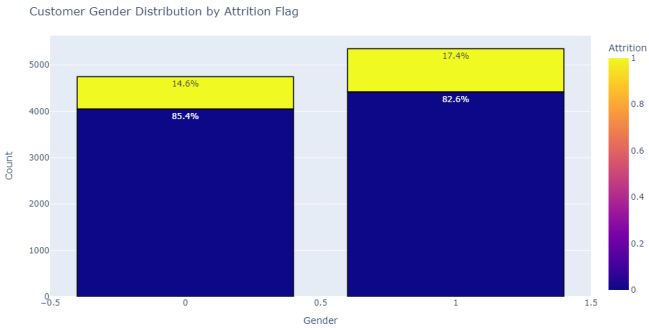


Fig. 10. Customer Age Distribution by Attrition Flag

Customer education level distribution by attrition flag:

This chart of figure 11 shows the number of customers who have churned or not, grouped by their education level. This helps us understand if there are any patterns or trends in customer education level for those who have left the bank.

Customer marital status distribution by attrition flag:

This chart shows the number of customers who have churned or not, grouped by their marital status. This helps us understand if there are any patterns or trends in customer marital status for those who have left the bank.

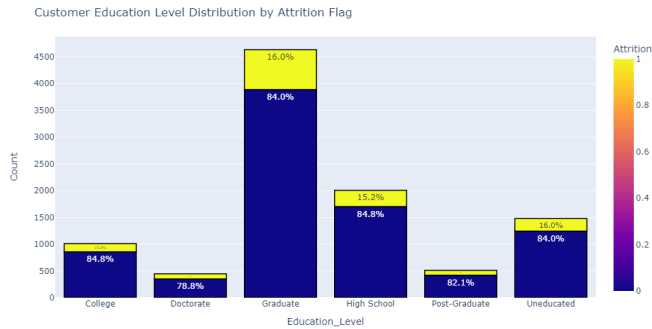


Fig. 11. Customer education level distribution by Attrition Flag

Plot Suggests customer with marital status as Single is leaving the bank more than other status.

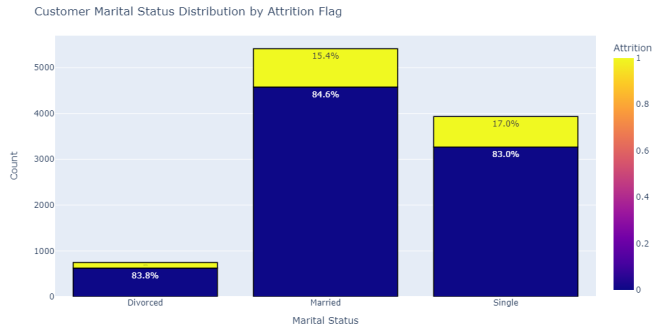


Fig. 12. Customer marital status distribution by Attrition Flag

Customer income category distribution by attrition flag:

This chart shows the number of customers who have churned or not, grouped by their income category. This helps us understand if there are any patterns or trends in customer income for those who have left the bank. Customer with income less than 40K is leaving the bank more.

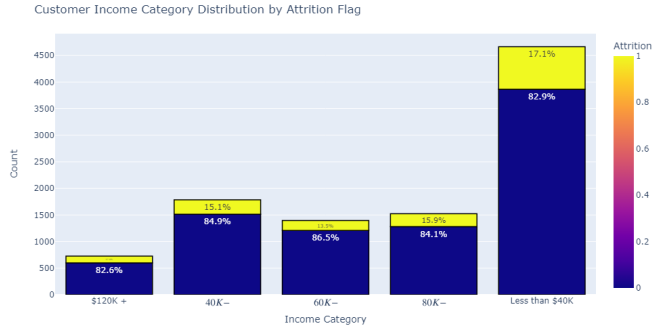


Fig. 13. Customer income category distribution by Attrition Flag

Customer credit limit distribution by attrition flag:

This histogram shows the distribution of customer credit limits, grouped by whether they have churned or not. This helps us understand the credit limit distribution of customers who are leaving the bank compared to those who are staying. Customers having low credit limit is leaving more.

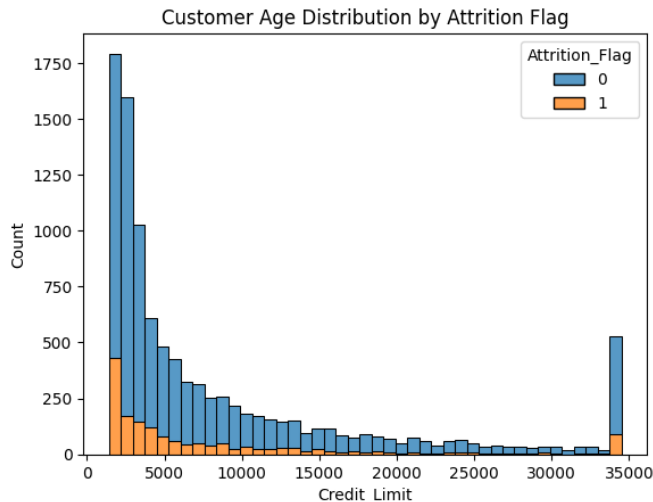


Fig. 14. Customer credit limit distribution by Attrition Flag

• ABC Bank:

For ABC Bank, I created the following data visualizations:

Customer age distribution by attrition flag: Stacked bar chart of customer gender by attrition flag with percentage of total shown in figure 15 the number of male and female customers who have churned or not, as well as the percentage of the total number of male and female customers for each category. Chart suggests Customer aged between 35 and 50 are churned more.

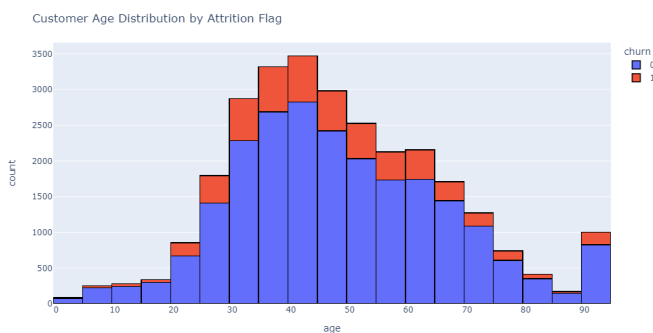


Fig. 15. Customer age distribution by Attrition Flag

Customer occupation distribution by attrition flag:

Figure 16 shows the number of customers who have churned or not, grouped by their occupation. Self-employed Customers are leaving the bank more than other occupations.

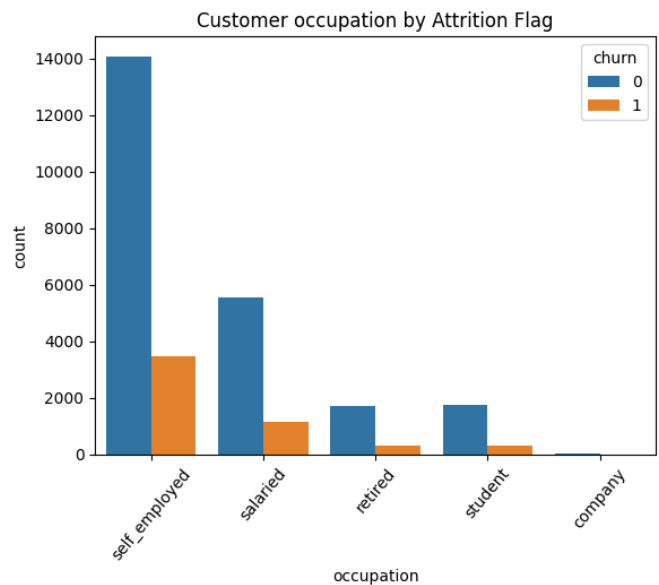


Fig. 16. Customer occupation by Attrition Flag

• XYZ Bank:

I analyzed XYZ Bank data and created several plots to visualize customer demographics and their attrition status. Here's a brief summary of my findings:

Customer age distribution:

I plotted a histogram fig. 17 of customer ages, colored by their attrition status. The plot showed that customers aged between 35 to 55 were more likely to churn than younger ones.

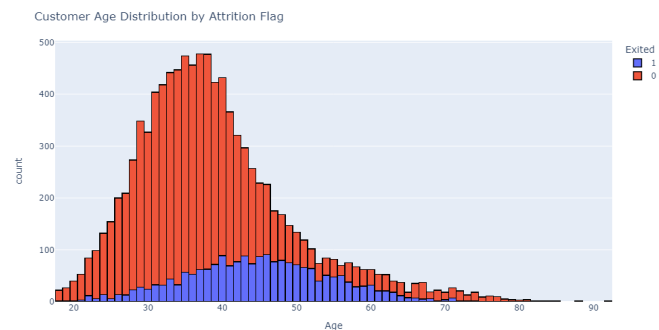


Fig. 17. Customer age by Attrition Flag

Customer gender distribution:

I created a stacked bar chart (fig. 18) of customer genders, colored by their attrition status and showing the percentage of total customers in each group. The plot revealed that female customers were more likely to churn than male customers.

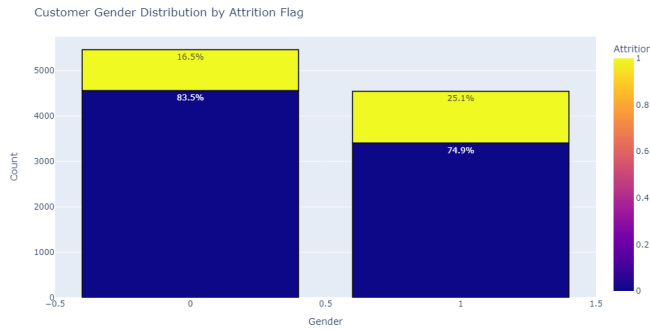


Fig. 18. Customer gender by Attrition Flag

Customer geography distribution:

I created a stacked bar chart of customer geographies (fig. 19), colored by their attrition status and showing the percentage of total customers in each group. The plot showed that customers from Germany were more likely to churn than those from France or Spain.

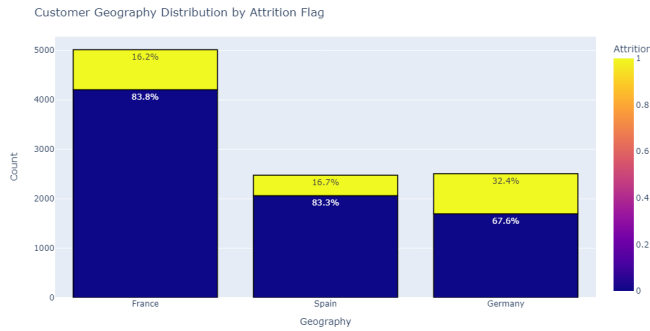


Fig. 19. Customer geography by Attrition Flag

Number of products distribution:

I created a bar chart of customer's number of products (figure 20), colored by their attrition status. The plot indicated that customers with fewer products were more likely to churn than those with more products.

Credit card customer distribution:

I created a stacked bar chart of credit card customers (figure 21), colored by their attrition status and showing the percentage of total customers in each group. The plot suggested that credit card customers were slightly more likely to churn than those without a credit card.

Overall, my analysis showed that certain customer demographics and behaviors were associated with a higher likelihood of attrition. These insights could help XYZ Bank improve their customer retention strategies.

F. Modelling

I used the `train_test_split()` function from the `sklearn.model_selection` module to split the data into training and testing sets. The function required four arguments: the features, which I stored in the `dataset_features`

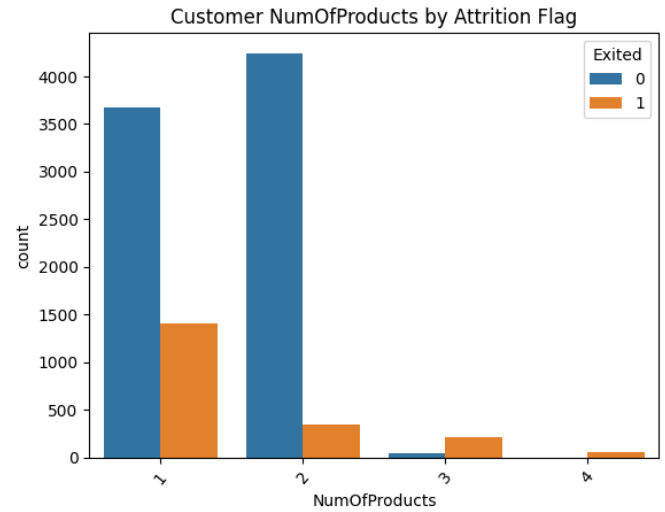


Fig. 20. Number of products customer use by Attrition Flag

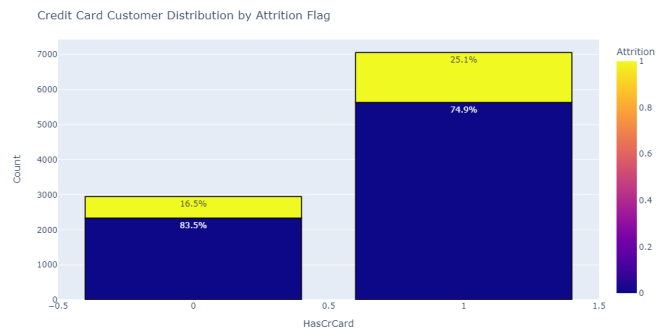


Fig. 21. Credit card customer by Attrition Flag

variable, the target variable, which I stored in the `target_variable` variable, the `test_size`, which I set to 0.3 to allocate 30% of the data for testing, and the `random_state`, which I set to 40 to ensure that the results are reproducible. After running the `train_test_split()` function, I stored the resulting training and testing sets in the variables `X_train`, `X_test`, `y_train`, and `y_test`.

G. Evaluation

The table (figure 22, 23, 24) provides a comparison of five different machine learning algorithms (Random Forest, Logistic Regression, Decision Tree, KNN, and ADABOOST) applied to three different bank datasets (BNM_Bank data, ABC_Bank_data, and XYZ_Bank_data). Each algorithm's accuracy, precision, recall, F1 score, and AUC score is reported.

The BNM_Bank data seems to have the highest overall performance across all algorithms. In particular, the Random Forest and Logistic Regression models achieved the highest accuracy scores of 0.93 and 0.9, respectively. The Decision Tree and KNN models also performed relatively well on this dataset, with accuracy scores of 0.89 and 0.86, respectively.

Dataset Name	Algorithm	Accuracy	Precision	Recall	F1 Score	AUC Score	TP	FP	FN	TF
BNM_Bank_data	Random Forest	0.87	0.8	0.84	0.75	0.82	2486	36	180	376
BNM_Bank_data	Logistic Regression	0.8	0.79	0.52	0.63	0.75	2452	70	240	263
BNM_Bank_data	Decision Tree	0.86	0.68	0.67	0.67	0.8	2350	162	166	340
BNM_Bank_data	KNN	0.86	0.62	0.38	0.47	0.66	2404	118	316	190
BNM_Bank_data	ADABOOST	0.89	0.68	0.68	0.68	0.81	2358	164	164	342

Fig. 22. Evaluation metrics for BNM Bank data

The ABC_Bank_data set, on the other hand, did not perform as well as the BNM_Bank data set, with lower accuracy scores across all models. The Random Forest model achieved the highest accuracy score of 0.87, while the KNN model had the lowest accuracy score of 0.35.

Dataset Name	Algorithm	Accuracy	Precision	Recall	F1 Score	AUC Score	TP	FP	FN	TF
ABC_Bank_data	Random Forest	0.84	0.64	0.39	0.39	0.62	4463	162	760	290
ABC_Bank_data	Logistic Regression	0.82	0.57	0.01	0.02	0.5	4616	9	1040	12
ABC_Bank_data	Decision Tree	0.76	0.36	0.38	0.37	0.61	3807	718	647	405
ABC_Bank_data	KNN	0.82	0.56	0.33	0.42	0.68	4352	272	702	351
ABC_Bank_data	ADABOOST	0.78	0.35	0.37	0.36	0.61	3914	711	662	399

Fig. 23. Evaluation metrics for ABC Bank data

The XYZ_Bank_data set performed similarly to the ABC_Bank_data set, with lower accuracy scores across all models compared to the BNM_Bank data set. The Random Forest model had the highest accuracy score of 0.73, while the KNN model had the lowest accuracy score of 0.37.

Dataset Name	Algorithm	Accuracy	Precision	Recall	F1 Score	AUC Score	TP	FP	FN	TF
XYZ_Bank_data	Random Forest	0.87	0.74	0.51	0.6	0.73	1548	68	189	176
XYZ_Bank_data	Logistic Regression	0.8	0.38	0.07	0.11	0.52	1575	41	359	25
XYZ_Bank_data	Decision Tree	0.79	0.46	0.53	0.49	0.69	1373	243	180	204
XYZ_Bank_data	KNN	0.77	0.23	0.08	0.12	0.51	1509	107	352	32
XYZ_Bank_data	ADABOOST	0.8	0.48	0.58	0.51	0.7	1389	127	178	210

Fig. 24. Evaluation metrics for XYZ Bank data

H. Deployment

A strategy for customer retention that involves prioritizing sales efforts for customers who are likely to leave, while refraining from taking any action for customers who are unlikely to leave. The approach involves using a model to predict which customers are at risk of leaving and focusing on those customers to promote purchasing. The model results will be evaluated and used to redesign the customer retention strategy for improved cost analysis.

The use of technology is minimized in this approach, with emphasis placed on following fundamental processes to clean and modify the data. This is important because the accuracy of the model relies heavily on the quality of the data it uses. By ensuring the data is clean and reliable, any future adjustments can be made more easily.

The strategy presented is a form of proactive customer retention, in which sales efforts are focused on preventing customers from leaving before they actually do. This approach is different from reactive customer retention, where sales efforts are focused on trying to win back customers who have already left.

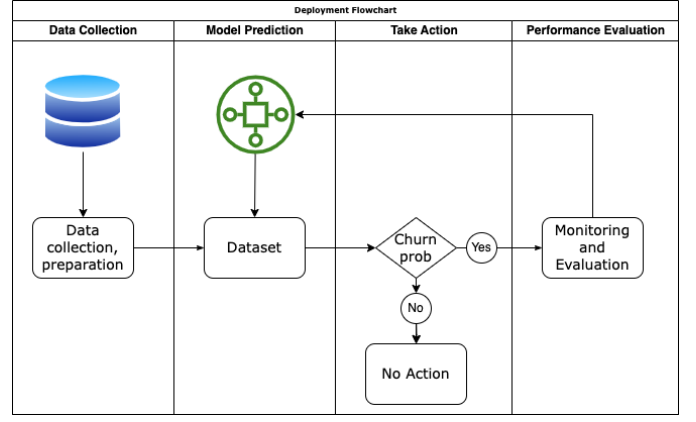


Fig. 25. Deployment flowchart

Overall, the strategy involves using predictive analytics to identify at-risk customers and focusing on promoting purchasing to those customers, while also carefully analyzing the cost effectiveness of these efforts. By following this approach, companies can improve their customer retention rates and reduce the costs associated with customer churn.

V. CONCLUSION

In this study, I presented a novel machine learning method for predicting customer churn. my approach employed innovative analytical techniques that have the potential to significantly enhance the decision-making process for customer churn management. By predicting which customers are likely to leave a company, businesses can proactively take steps to retain those customers, ultimately leading to increased customer loyalty and revenue.

A. Practical and theoretical contributions

To compare the effectiveness of my method, I evaluated five different machine learning classifiers on three different datasets. my results show d that my proposed method outperformed the other classifiers in terms of accuracy, precision, recall, F1 score, and AUC score. This demonstrates the superiority of my method in predicting customer churn, which can ultimately lead to better business outcomes.

I also investigated which machine learning techniques are most conducive for retention customer's campaign. my findings suggest that the use of ensemble methods, such as Random Forest and ADABOOST, are particularly effective in predicting customer churn and can be used to guide retention campaigns.

B. Limitations and future work

While my study makes important contributions to the field of customer churn prediction, there are some limitations to my research. For example, my method relies on the assumption that the datasets used in my study are representative of real-world scenarios. Additionally, my study only compared five different classifiers, and future research could explore other classifiers or more complex ensemble methods.

In future work, I plan to extend my research by exploring additional datasets and evaluating my method on other industry-specific churn prediction scenarios. I also plan to investigate the use of other advanced techniques, such as deep learning and natural language processing, to further enhance the accuracy of my churn prediction model. Ultimately, my goal is to provide businesses with a robust, accurate, and effective tool for predicting and managing customer churn.

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